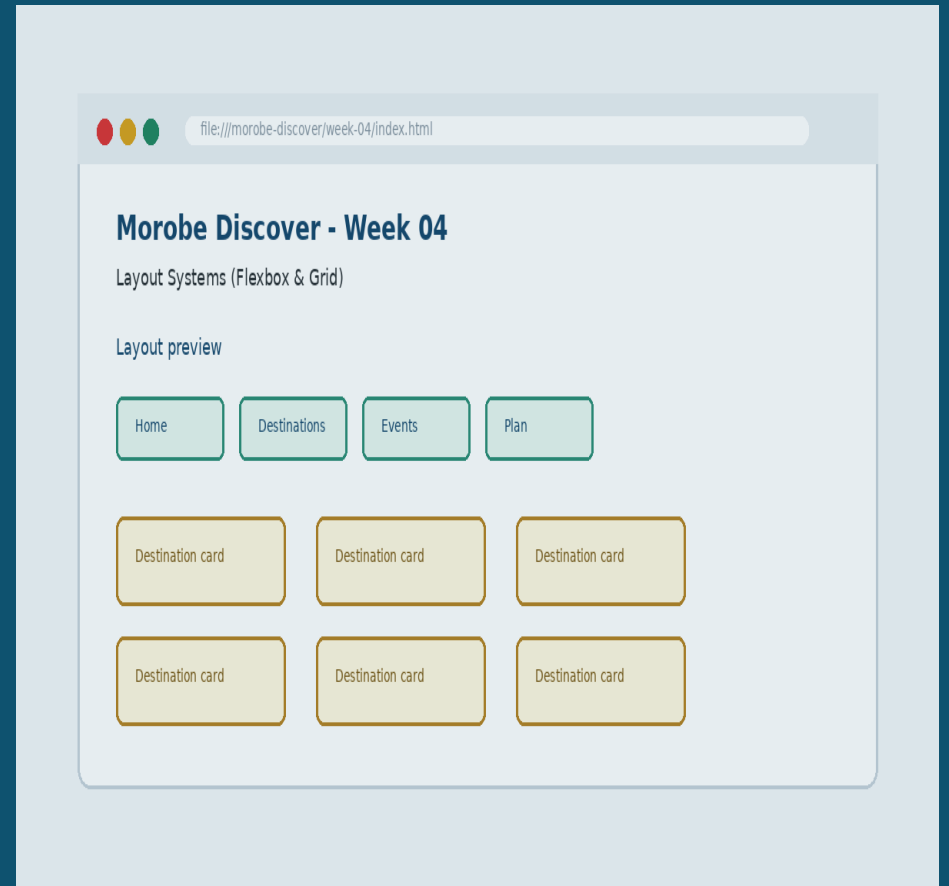


Week 04

Layout Systems (Flexbox & Grid)

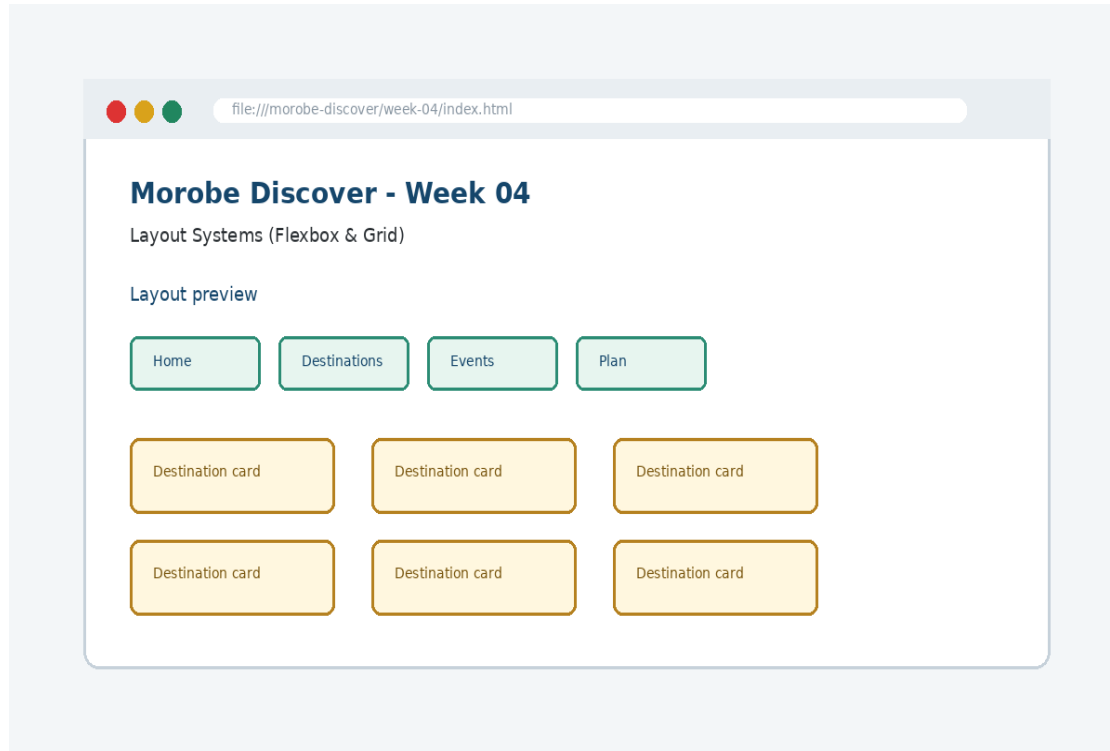
Manage one-dimensional and two-dimensional layouts for complex page sections.



Learning outcomes

Manage one-dimensional and two-dimensional layouts for complex page sections.
Explain the core programming/design concept for this week
Read and modify key code patterns
Test the weekly project increment
Prepare evidence for GitHub submission

Today's problem



Why it matters

Use Flexbox for one-dimensional alignment and Grid for two-dimensional page and card structures.

Project before -> after

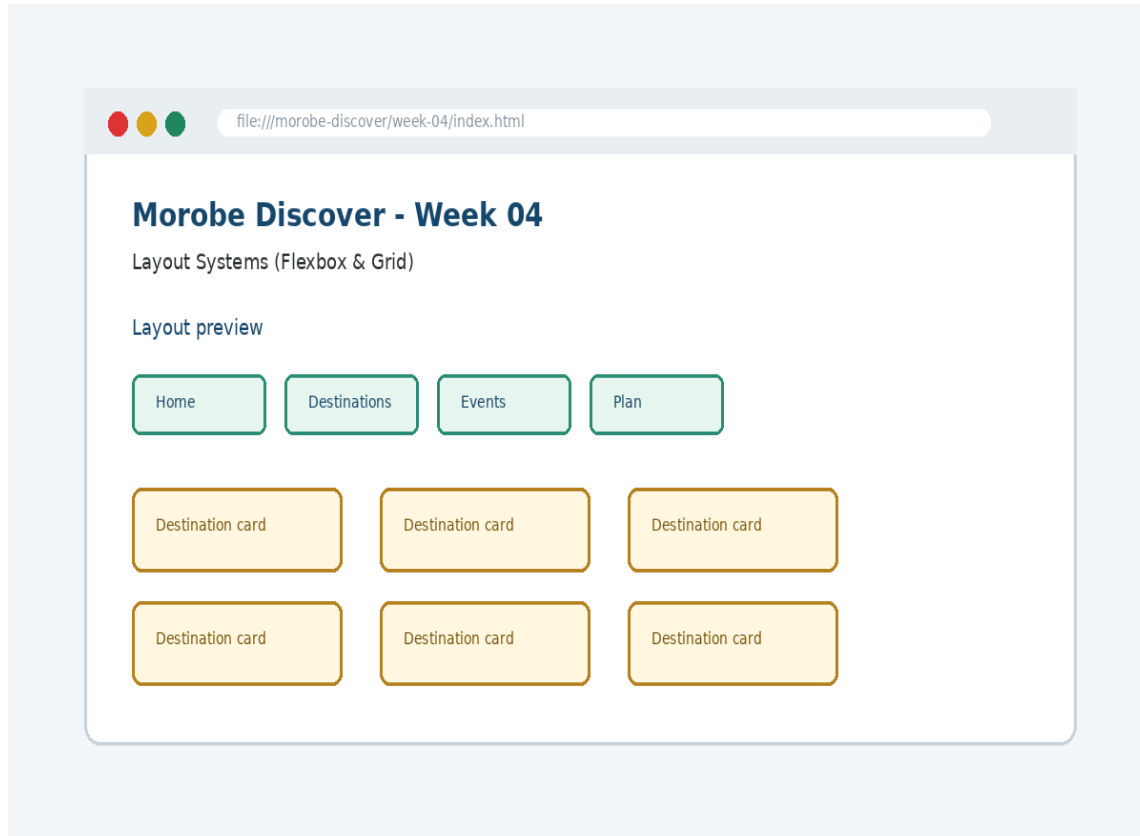
Before Week 04

Week 03 produced forms and a basic visual design system.

After Week 04

The website uses intentional layout systems for navigation, forms, cards and page sections.

What we build this week



Convert navigation into a Flexbox row
Align form controls with consistent gap
Create a card gallery with CSS Grid
Use container width to improve readability
Remove manual spacing hacks

Core principle

Use Flexbox for one-dimensional alignment and Grid for two-dimensional page and card structures.

Keep in mind

A good web developer can point to the code, the browser result, and the user reason behind the decision.

Key concepts

Concept 1

Flexbox aligns items along one main axis

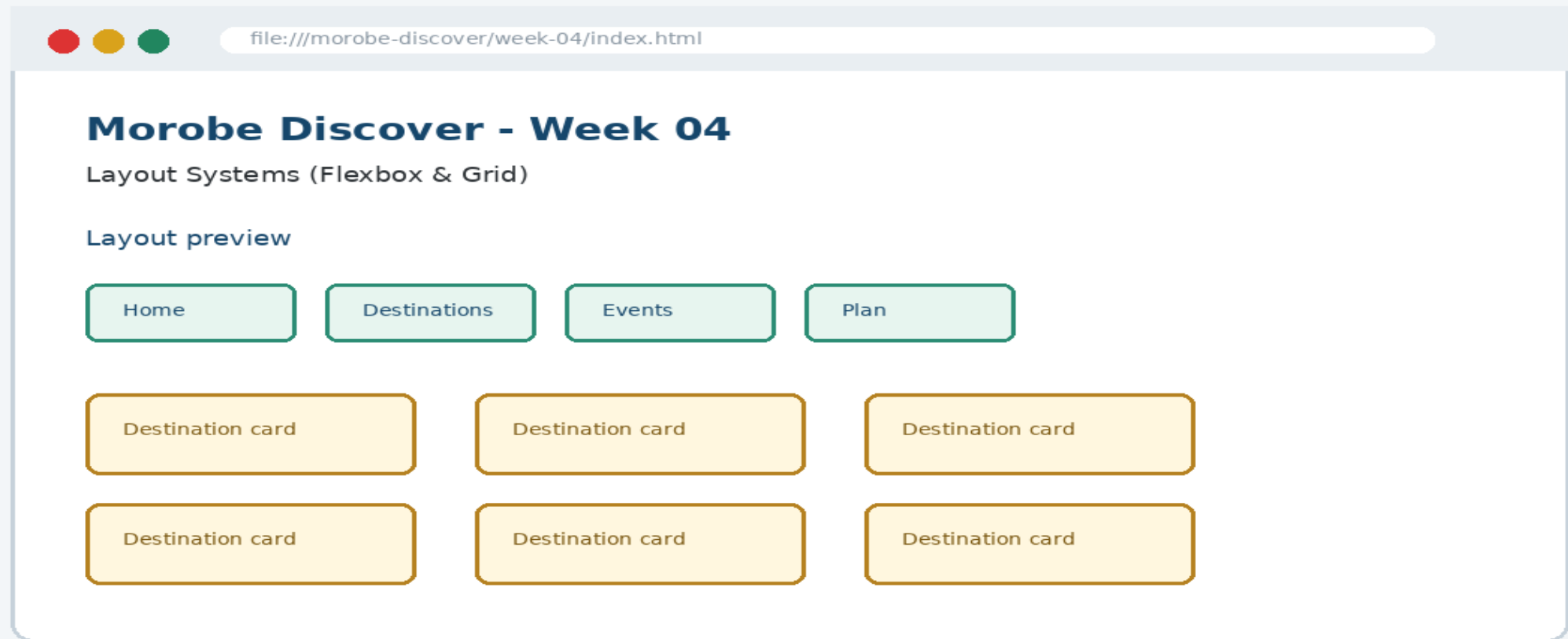
Concept 2

Grid creates rows and columns together

Concept 3

Gap creates maintainable spacing without layout hacks

Project anatomy



Key code pattern A

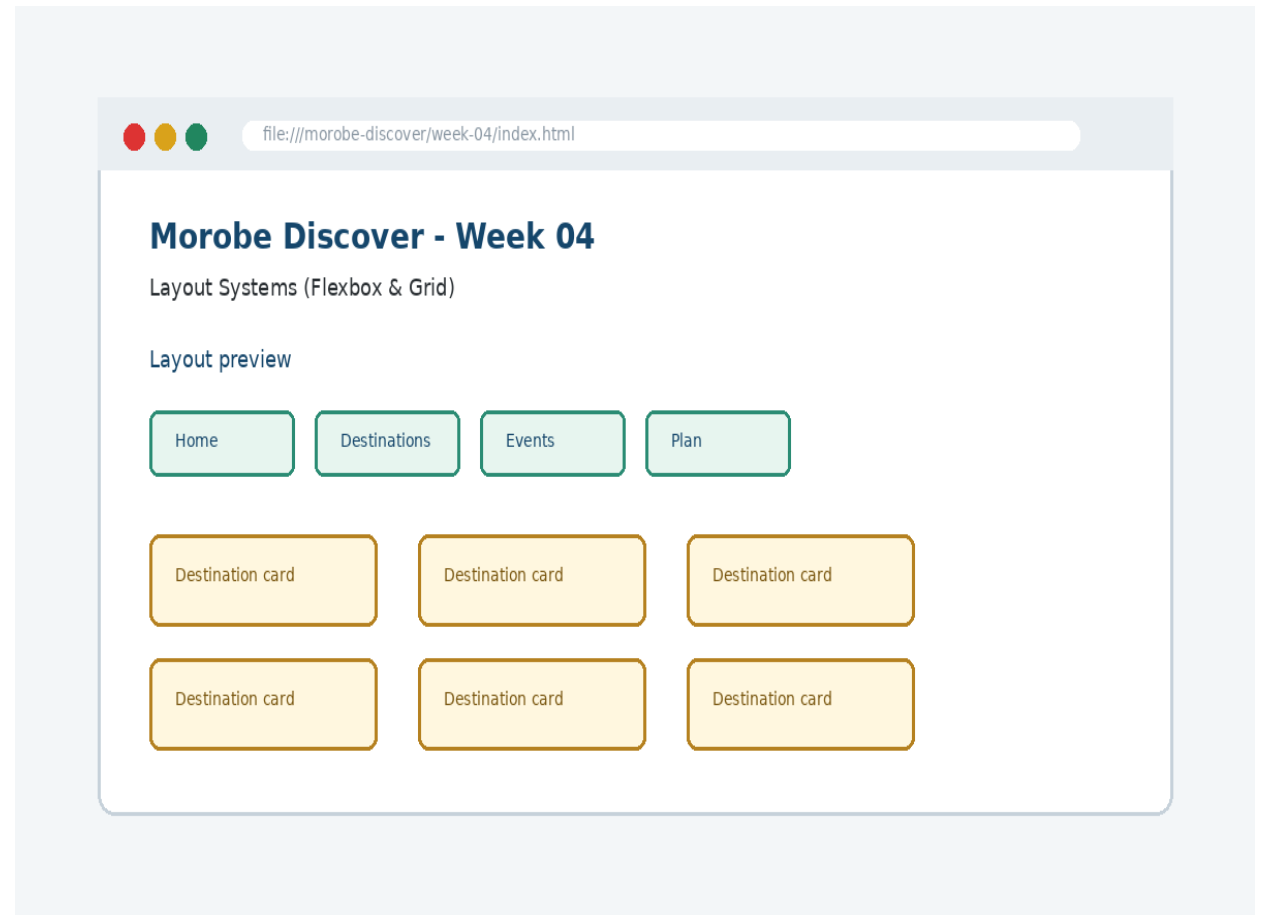
```
.site-nav {  
  display: flex;  
  gap: 1rem;  
  align-items: center;  
}
```

Explain it

Which file contains this code?
Which line creates the visible change?
What should appear in the browser?

Code -> visual result

```
.site-nav {  
  display: flex;  
  gap: 1rem;  
  align-items: center;  
}
```



What changed?

Project file changed for Layout Systems (Flexbox & Grid)
The browser output is now closer to the required weekly evidence
The change supports a user task or quality requirement
The change can be tested and committed

Key code pattern B

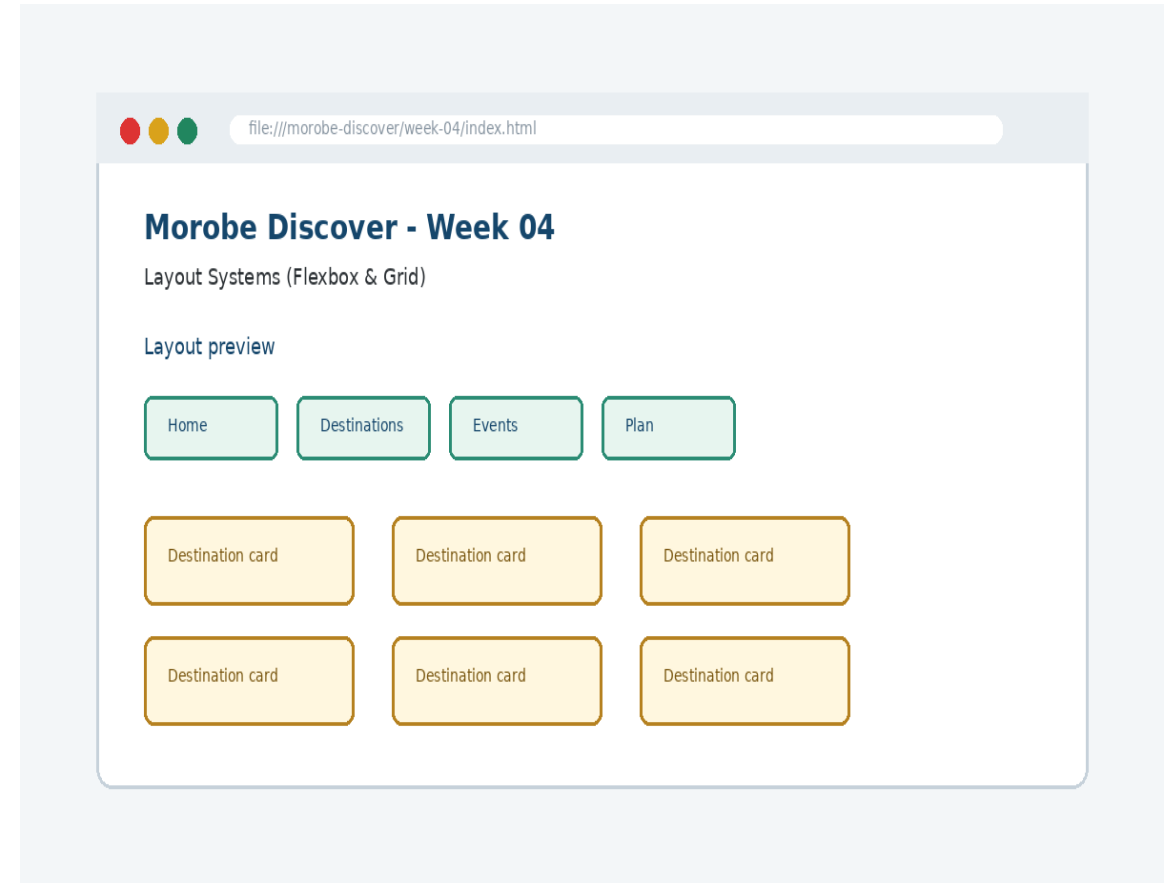
```
.card-grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 1.25rem;  
}
```

Why this pattern is needed

The second pattern usually applies the same idea to a different component, state, layout or data flow.

Apply it to Morobe Discover

Convert navigation into a Flexbox row
Align form controls with consistent gap
Create a card gallery with CSS Grid
Use container width to improve readability



Compare the approaches

Weak approach

Memorise the definition
Copy code without understanding
No browser evidence
No test note

Professional approach

Explain the concept
Apply the correct code pattern
Inspect the result
Commit and describe the evidence

Common mistake

Mistake

The page appears unchanged, but the student assumes the feature is complete.

Correction

Check the correct file, refresh the browser, inspect DevTools, and verify the expected output.

Why does this fail?

```
// Broken checkpoint  
// File changed, but the browser result does not  
change.  
// Possible causes:  
// - wrong file  
// - file not linked  
// - syntax error  
// - browser not refreshed
```

Class question

Which cause would you check first, and why?

Fix the problem

- Confirm the browser is opening the correct file
- Check links to CSS/JS/data files
- Inspect the console for errors
- Use a small working example
- Test again before committing

Class activity

Activity - 7 minutes

Choose Flexbox or Grid for navigation, hero area, card gallery and form rows.

Work in pairs

Point to the project evidence

Prepare one answer to share

Implementation steps 1-3

Convert navigation into a Flexbox row
Align form controls with consistent gap
Create a card gallery with CSS Grid

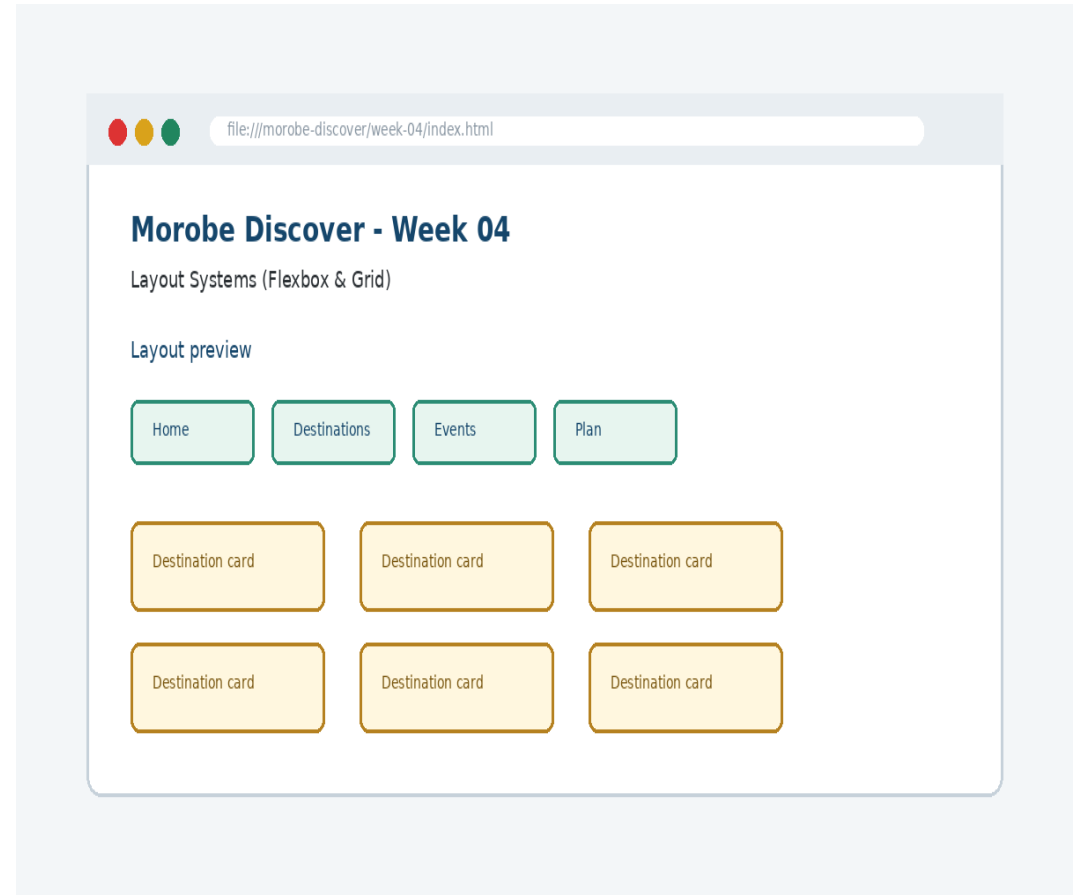
Implementation steps 4-5

Use container width to improve readability
Remove manual spacing hacks

LIVE DEMO

Demo checklist

1. Open the current project
2. Locate the affected file
3. Add or modify the key code
4. Refresh the browser
5. Inspect the result
6. Explain the change



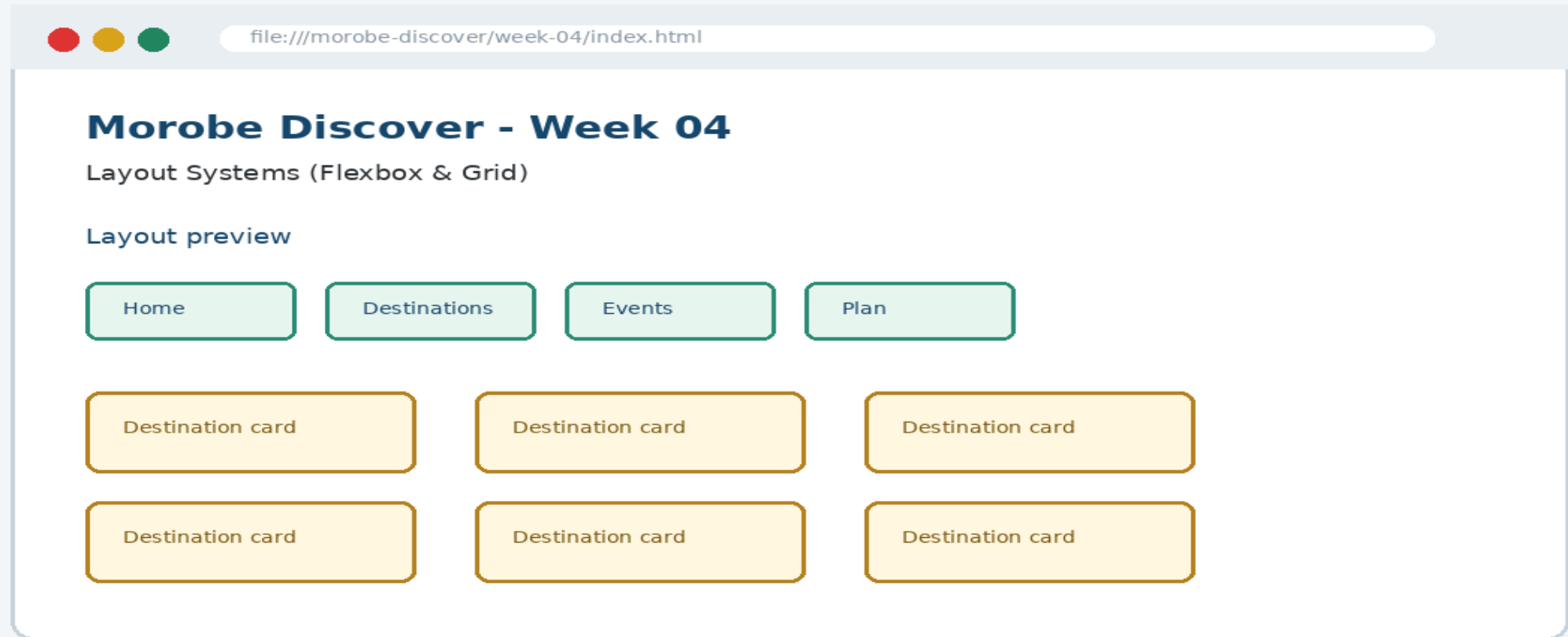
Predict the output

```
.card-grid {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  gap: 1.25rem;  
}
```

Prediction prompt

What should change on screen when this code is applied correctly?

Browser result



Testing the feature

- Resize the browser width
- Check navigation alignment
- Inspect card gaps and wrapping
- Confirm form labels align with controls

Inspect with browser tools

Inspection path

Right click -> Inspect
Elements panel -> structure/classes
Console -> JavaScript errors
Network/Application -> data/storage when relevant

Evidence to save

Screenshot
Short test note
Changed files
Git commit message

Git evidence

```
git add .  
git commit -m "Week 04 Layout Systems (Flexbox &  
Grid) project increment"
```

Why it matters

Git history shows progression from Week 01 to final deployment. It is part of the assessment evidence.

Complete source code

Repository:
`https://github.com/YOUR-USERNAME/is229-morobe-discover`

Weekly folder:
`/week-04/`

Complete project:
`/complete-project/`

Reminder

Do not paste a whole file into a slide. Use the repository for complete code.

Mini challenge

Challenge

Modify one part of the Week 04 implementation and explain what changed in the browser.

State the file edited

State the line/section changed

Show the before and after result

What counts as evidence?

Technical evidence

Changed code files
Browser screenshot
Test note
Git commit

Understanding evidence

Code explanation
Why this method fits
Connection to previous week
Connection to next week

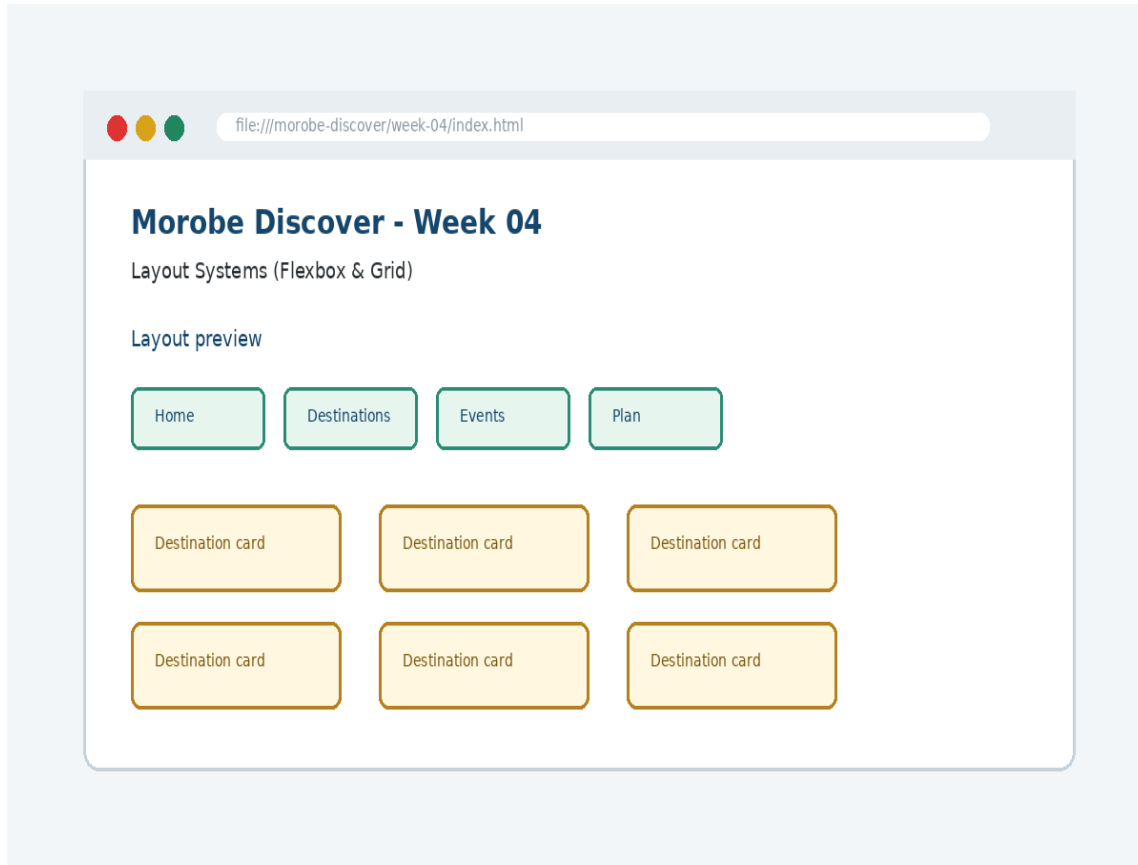
Project lab handoff

- Convert navigation into a Flexbox row
- Align form controls with consistent gap
- Create a card gallery with CSS Grid
- Use container width to improve readability
- Remove manual spacing hacks

Definition of done

Week 04 feature is implemented
Project still preserves earlier week features
Browser result matches the expected output
Test note and screenshot are saved
Git commit message is meaningful

What we built today



The website uses intentional layout systems for navigation, forms, cards and page sections.

Evidence: Navigation bar, button rows, aligned form controls and card galleries

Technology: CSS Flexbox / CSS Grid

Lecture summary

Use Flexbox for one-dimensional alignment and Grid for two-dimensional page and card structures.

Flexbox aligns items along one main axis

Grid creates rows and columns together

Testing and Git are part of development, not extras

Prepare for next class

Finish this week's lab before the next class
Read the next topic in advance
Bring the current project ready to extend
Be ready to explain one code decision

Exit question

How did Week 04 improve Morobe Discover, and what will the next week build on?